SilkSphere: YOLOv8 Silkworm Disease Detection

Model comparison report: Original vs Tray-Optimized YOLOv8s | 14 October 2026 | Himanshu Chib

99.1%	**+6.0 pts**	**12.3 vs 14.1 ms**	**93% fewer**
Tray model mAP50 (original: 94.2%)	Precision gain (91.5% to 97.5%)	Inference time original vs tray	Training images (500 vs 7,576)

Executive summary

Two YOLOv8s detectors classify silkworms as **Grasserie (infected)** or **Healthy**. The **original model** was trained on 7,576 real close-up photographs of single worms. The **tray-optimized model** was trained on 500 synthetic tray scenes (collages of real worm images) holding many small worms. The tray model is more accurate on every metric (+4.9 points mAP50) and costs 1.8 ms more per image.
Recommendation: use the tray model for the project review and tray or collage scenes; keep the original model for real-farm deployment.

Models compared

	Original	Tray-optimized
Training data	Real close-ups	Synthetic tray scenes
Train / val images	7,576 / 1,000	500 / 100
Epochs / batch	100 / 16	50 (early stop) / 8
Training time	~3 hours	~25 minutes
Augmentation	Flip, rotate, scale, HSV	Mosaic, flip, rotate, scale, HSV
Target scene	1 large worm per image	~20-25 small worms per tray
Architecture	YOLOv8s, 11.1M params, 22.5 MB, 28.4 GFLOPs	same

Key metrics

Metric	Original	Tray	Change
mAP50	94.2%	99.1%	+4.9
mAP50-95	85.1%	89.8%	+4.7
Precision	91.5%	97.5%	+6.0
Recall	92.8%	97.1%	+4.3
Grasserie P / R	91 / 93%	96.3 / 95.5%	+5.3 / +2.5
Healthy P / R	92 / 93%	98.6 / 98.7%	+6.6 / +5.7
Inference	12.3 ms (81 FPS)	14.1 ms (71 FPS)	+1.8 ms

Accuracy: overall and per class

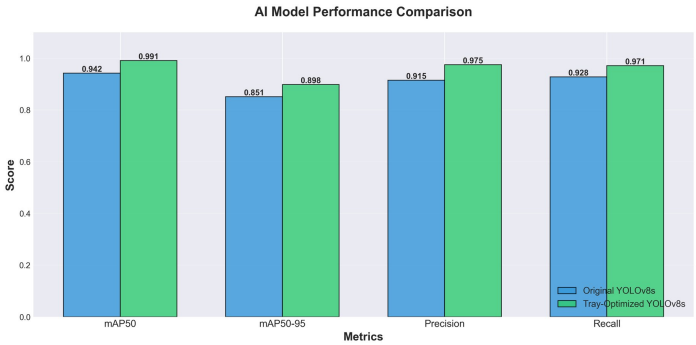


Figure 1. Overall metrics



Figure 2. Per-class precision and recall

Speed vs accuracy

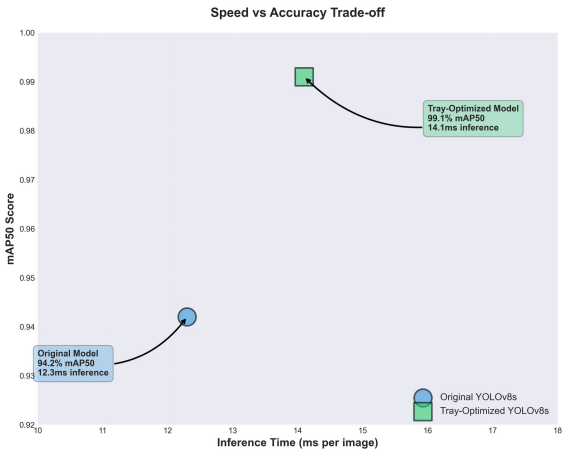


Figure 3. Inference time vs mAP50

The original model is faster (12.3 ms, 81 FPS) than the tray model (14.1 ms, 71 FPS), a **14.6% slowdown**, while the tray model gains **4.9 points of mAP50**. Model size is identical (22.5 MB).

Within the tray model, Healthy (98.6% / 98.7%) is detected more reliably than Grasserie (96.3% / 95.5%), so infected worms remain the class to watch.

Minimum hardware: NVIDIA RTX 3050 or better, 4 GB RAM, 2 GB VRAM. Software: Python 3.10, PyTorch 2.5.1, Ultralytics YOLOv8, CUDA 12.1.

Training data and use cases

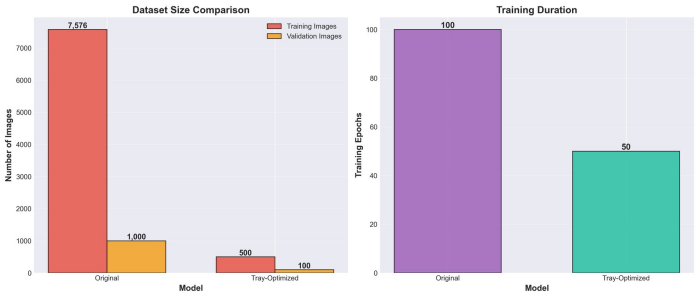


Figure 4. Dataset size and training epochs



Figure 5. Intended use cases

The tray model reached higher scores with 93% fewer images and half the epochs, because it was trained and tested on the same kind of scene it is used for.

Independent evaluation run (88 epochs, Roboflow dataset, 2 classes)

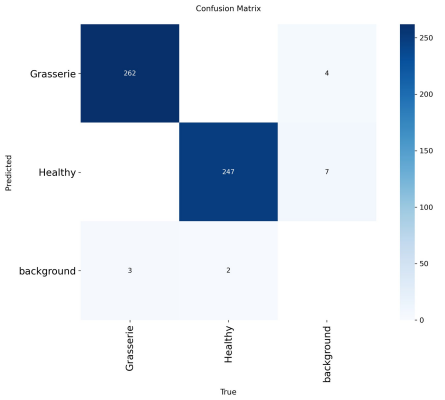


Figure . Confusion matrix

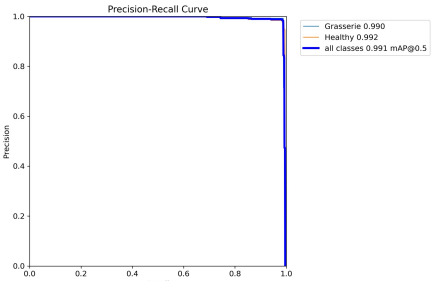


Figure 7. Precision-recall curve

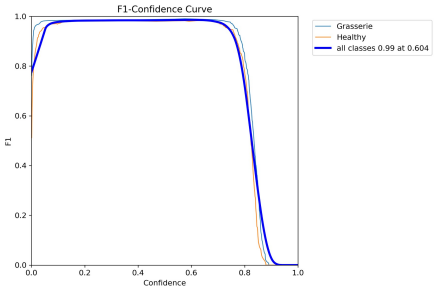


Figure 8. F1-confidence curve

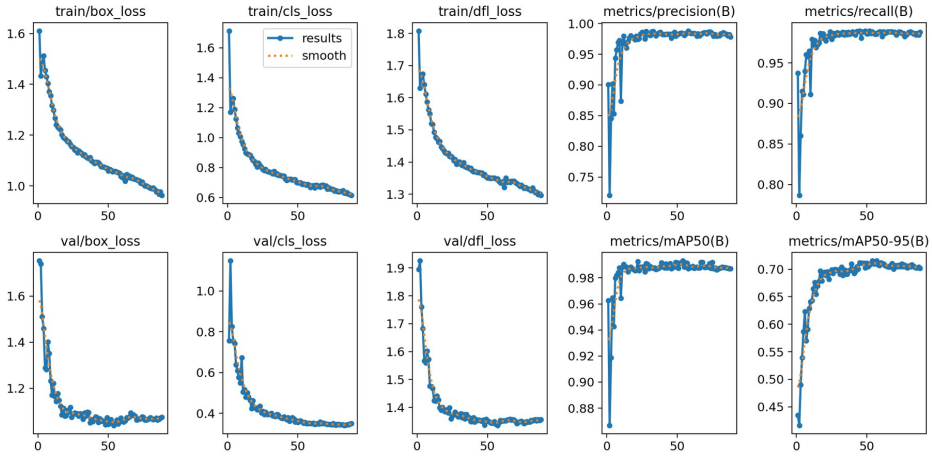


Figure 9. Training curves (88 epochs)

mAP50 0.991 (Grasserie 0.990, Healthy 0.992); precision about 0.98, recall about 0.99; mAP50-95 about 0.70.

Validation: 262 Grasserie and 247 Healthy detected correctly, **no Grasserie/Healthy confusions**; losses fall and flatten with no sign of overfitting.

Best F1 0.99 at confidence 0.604; precision reaches 1.00 at 0.855; recall drops sharply above about 0.75. **Operating threshold: 0.5-0.6.**

Recommendation

Use case	Model	Reason
Project review, demo, printed cutouts, tray and collage scenes	Tray-optimized	99.1% mAP50; better small-object detection on multi-worm scenes
Real farm deployment, IoT monitoring	Original	Trained on 7,576 real images; faster (12.3 ms, 81 FPS)

Next steps: test both models on more real tray images; collect real tray data for retraining; evaluate an ensemble; deploy the original model in the production IoT system.

Notes before presenting

- The two models were validated on different sets (1,000 real close-ups vs 100 synthetic tray scenes), so the 4.9-point gap is not a like-for-like comparison and may overstate real-world tray performance.
- The independent evaluation shows 88 epochs and mAP50-95 of about 0.70, while the comparison lists 50 epochs and 89.8% for the tray model. Confirm which run each figure belongs to.